TEXORA · LECTURE SHEET

Cost Calculation

Building the FOB price of a garment: fabric, trims, cut-and-make (CM), testing and overhead assembled into a costing sheet, with the profit margin applied last — worked for the course shirt order.

AM-3207 · ADVANCED MACHINERIES AND RELEVANT CALCULATIONS · TOPIC 6 · SPECIALIZATION — APPAREL MANUFACTURING

1. The anatomy of a garment price

An export garment price (FOB — free on board) is a sum of calculated parts, each owned by an earlier topic of this course: **fabric** (Topic 5 consumption × fabric price), **trims and accessories** (Topic 4), **CM** — the cut-and-make charge covering cutting, sewing and finishing labour, factory running cost and the maker's margin on conversion — plus **process additions** where they apply (washing, AM-4101 Topic 5; printing, AM-4101 Topic 3), **testing and inspection**, and **overheads**. Profit is applied to the assembled cost at the end. A costing sheet is therefore an audit trail: challenge any line and its calculation answers.

2. The costing sheet — course shirt

COST ELEMENT	BASIS OF CALCULATION	US$ / PIECE	SHARE
Fabric	0.927 m (gross, Topic 5) × $3.20/m	2.97	69.4%
Trims & accessories	Thread, buttons, labels, interlining, polybag, carton share (Topic 4)	0.40	9.4%
CM (cut & make)	Cutting, sewing, finishing conversion charge	0.63	14.6%
Testing & inspection	Fabric/garment tests and third-party final inspection, amortised	0.10	2.3%
Overhead & commercial	Banking, documentation, transport to port, miscellaneous	0.18	4.2%
Total cost		**4.27**	100%
Profit margin 10%	10% on total cost	0.43	

FOB price	**4.70 / piece · 56.38 / dozen**

Figures are rounded to cents for presentation; the unrounded fabric line is 0.927 × 3.20 = $2.9664, total cost $4.2714, FOB $4.69854 — quoted to the buyer as $4.70 per piece. For the full order the FOB value is 12,000 × 4.70 = **$56,400** (approximately, at the quoted price).

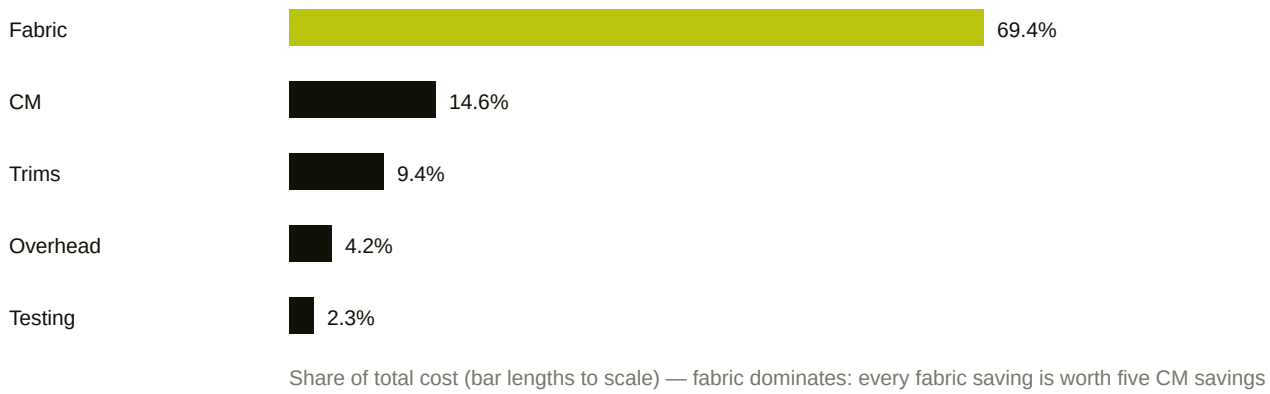


Figure 1 — Cost structure of the course shirt. The dominance of fabric (about 70%) is typical of woven garments and is why Topics 3 and 5 precede this one.

3. Where the CM figure comes from

CM is itself a calculation. A factory derives its conversion cost per minute from its monthly running cost divided by its available machine-minutes, then prices a style as SMV × cost per minute (SMV is built in Topic 7). Quoted CM also absorbs style risk: short orders, complex operations and new-buyer styles carry higher CM because set-up and learning losses are spread over fewer pieces. When a buyer negotiates, fabric and CM are negotiated differently — fabric is a pass-through of a market price times a verifiable consumption, while CM is the factory's own efficiency made visible.

4. Costing discipline

- **Cost at gross consumption, always.** Net figures flatter the price and bankrupt the order.
- **One currency, stated exchange basis.** Mixed Tk/$ sheets hide errors; convert once, at a stated rate.
- **Every addition visible.** Washing, printing, embroidery and special packing appear as their own lines (AM-4101 Topic 5 costs a wash at $4.50 per dozen), never buried inside CM.
- **Re-cost after any change.** A fabric width change alters consumption (Topic 3), hence fabric cost, hence FOB — the sheet is re-issued, not patched verbally.

Glossary

TERM	MEANING
FOB	Free on board: price including all costs until the goods are loaded on the vessel at the named port.
CM	Cut-and-make charge: the conversion price for cutting, sewing and finishing a garment.
Costing sheet	The itemised build-up of a garment's price from consumption and rate lines.
Cost per minute	A factory's running cost divided by available machine-minutes; the basis of CM calculation from SMV.

Key points and summary

- FOB = fabric + trims + CM + process additions + testing + overhead, with profit applied to the total.
- Course shirt: cost $4.27/pc, FOB $4.70/pc ($56.38/dozen); fabric is 69.4% of cost.
- Each line traces to an earlier calculation — consumption, trims, SMV — so the sheet can be defended line by line.
- Process additions are separate lines: the wash costed in AM-4101 adds $0.375 to each washed garment.

Practice

Answer in your own words; brief model answers follow.

1. Fabric price rises to $3.60/m with consumption unchanged. Recalculate the fabric line, total cost and FOB per piece at the same 10% margin.
2. A buyer demands a 5% price cut. Using the cost structure, show the cut in dollars per piece, and state which line can least absorb it and why.
3. The shirt is given an enzyme wash costed at $4.50 per dozen (AM-4101 Topic 5). What is the new FOB per piece if the margin is applied to the higher cost?
4. Why is profit applied after overheads rather than on the fabric-plus-CM subtotal?

Answer key

1. Fabric = 0.927 × 3.60 = $3.3372 ≈ $3.34. Total cost = 3.3372 + 0.40 + 0.63 + 0.10 + 0.18 = $4.6472 ≈ $4.65. FOB = 4.6472 × 1.10 = $5.11/pc.
2. 5% of $4.70 = $0.235/pc. Fabric can least absorb it: fabric is a bought-in market price times a fixed consumption — the factory does not control it — so the cut falls on CM and margin, the factory's own money.

3. Wash per piece = 4.50 ÷ 12 = $0.375. New cost = 4.2714 + 0.375 = $4.6464; FOB = × 1.10 = $5.11/pc.
4. Overheads are real costs of executing the order (banking, documentation, transport). Margin earned on only part of the cost understates the return the business requires on the whole commitment.
